Exploring Altermagnetism

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What is Altermagnetism?

Recently, a nonrelativistic spin-symmetry classification and description, focusing within the hierarchy of interactions on the strong Coulomb (exchange) interaction, has divided all collinear magnets into three mutually exclusive spin-group classes[1]:

1. Conventional ferromagnetic with strong (exchange) $\mathcal{T}$ reversal symmetry breaking in the electronic band structure and net magnetization
2. Conventional antiferromagnetic with, at least in the nonrelativistic limit, $\mathcal{T}$ reversal symmetric electronic band structure and zero net magnetization, and
3. A third class dubbed altermagnetic with strong (exchange) $\mathcal{T}$ reversal symmetry breaking in the electronic band structure and with, at least in the nonrelativistic limit, zero net magnetization.

[1] Olena Fedchenko et al. "Observation of time-reversal symmetry breaking in the band structure of altermagnetic RuO_2". In: *Science Advances* 10.5 (2024), eadj4883. DOI: 10.1126/sciadv.adj4883

What is Altermagnetism?

Or simply put, altermagnets are magnetic materials with no net magnetization but a spin-split electronic structure, combining traits of both antiferromagnets and ferromagnets, with unique properties useful in spintronics.

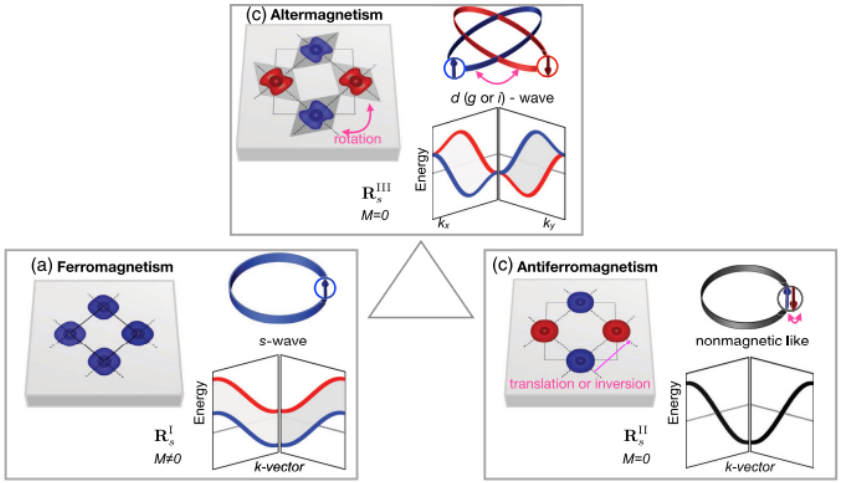


Figure: Comparison of collinear ferromagnetism, antiferromagnetism, and altermagnetism showing distinct spin configurations and electronic structures.[2]

[2] Libor Šmejkal, Jairo Sinova, and Tomas Jungwirth. "Emerging Research Landscape of Altermagnetism". In: *Phys. Rev. X* 12 (4 Dec. 2022), p. 040501. DOI: 10.1103/PhysRevX.12.040501

Magnetism and Superconductivity

In condensed-matter physics, altermagnetism enables unconventional d-wave magnetism, akin to d-wave superconductivity. Conventional s-wave superconductivity and magnetism involve electron-electron Cooper pairs and majority spin electron—minority spin-hole pairs, respectively. The discovery of d-wave superconductivity sparked interest in finding a similar d-wave phenomenon in magnetism.

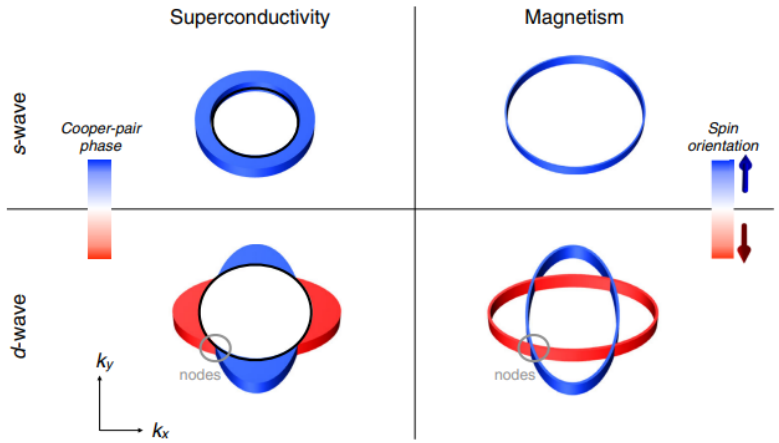


Figure: Analogy between s-wave and d-wave phases in superconductivity and magnetism, highlighting sign changes and nodes around the Fermi surface.[2]

Current Research

Altermagnetism was theoretically predicted[3] in 2021, proposing a new class of magnetic materials with distinct spin-split electronic structures but no net magnetization. It was experimentally verified in 2024 using materials such as manganese telluride[4] (MnTe) and ruthenium dioxide[1] (RuO_2), where researchers confirmed the characteristic properties using spin- and angle-resolved photoemission spectroscopy (ARPES).

[3] Libor Šmejkal, Jairo Sinova, and Tomas Jungwirth. "Beyond Conventional Ferromagnetism and Antiferromagnetism: A Phase with Nonrelativistic Spin and Crystal Rotation Symmetry". In: *Physical Review X* 12.3 (Sept. 2022). ISSN: 2160-3308. DOI: 10.1103/physrevx.12.031042

[4] J. Krempaský et al. "Altermagnetic lifting of Kramers spin degeneracy". In: *Nature* 626.7999 (Feb. 2024), pp. 517–522. ISSN: 1476-4687. DOI: 10.1038/s41586-023-06907-7

Thin Film (RuO_2)

"We have grown epitaxial $RuO_2(110)$ films with a thickness of 34 nm by pulsed laser deposition on $TiO_2(110)$ substrates that were heated during deposition to 400°C. The samples show no detectable remanent magnetization, consistent with the earlier magnetometry studies of analogous RuO_2/TiO_2 thin films."[1]

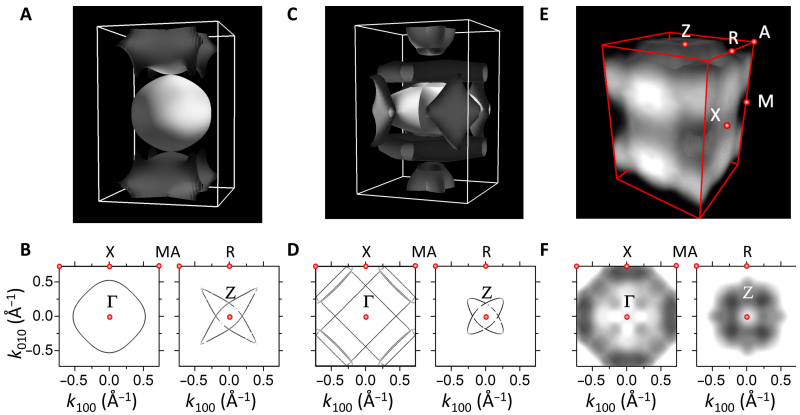


Figure: Fermi surface analysis of RuO : (A) Calculated for paramagnetic phase, (B) Cut at Γ-X-M and Z-A-R planes, (C) Calculated for collinear magnetically ordered phase, (D) Cut for altermagnetic phase, (E) Experimental topographic mapping, (F) Photoelectron intensity for planes as in (B).[1]

Thin Film (MnTe)

"MnTe(0001) epilayers of 200nm thickness were grown by molecular-beam epitaxy on single-crystalline In-terminated InP(111) substrates using elemental Mn and Te sources. The less than 1% lattice mismatch results in single-crystalline hexagonal MnTe growth with the c axis (z axis) perpendicular to the surface. Two-dimensional growth of α-MnTe is achieved at substrate temperatures of 370-450°C."[4]

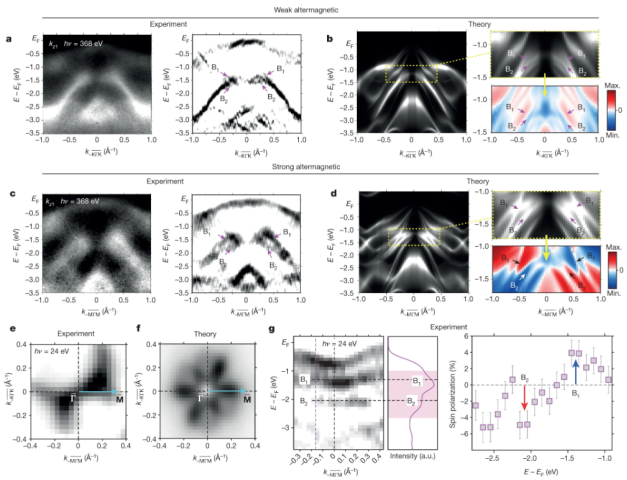


Figure: Experimental and theoretical ARPES band maps showing weak and strong altermagnetism along different momentum paths in MnTe.[4]

Identification

Elementary rules for identifying the altermagnetic phase of a crystal can be summarized as follows[2]:

1. There is an even number of magnetic atoms in the unit cell, and the number of atoms in the unit cell does not have to change between the nonmagnetic and magnetic phases of the crystal.
2. There is no inversion center between the sites occupied by the magnetic atoms from the opposite-spin sublattices.
3. The two opposite-spin sublattices are connected by crystallographic rotation transformation, possibly combined with translation or inversion transformation.
4. The spin group is determined by the algorithm described in Libor Šmejkal, Jairo Sinova, and Tomas Jungwirth. "Emerging Research Landscape of Altermagnetism". In: *Phys. Rev. X* 12 (4 Dec. 2022), p. 040501. DOI: 10.1103/PhysRevX.12.040501

Identification - Example

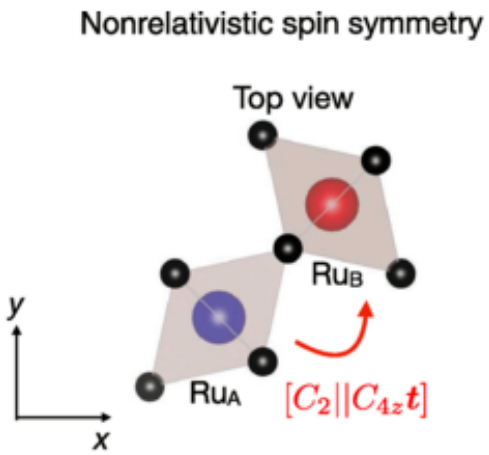


Figure: Schematic of RuO_2 crystal with opposite spin directions on Ru_A and Ru_B sublattices, highlighting spin-group symmetries and sublattice transformation generators.[2]

1. Two Ru atoms in the RuO_2 unit cell.
2. The absence of the inversion center between the Ru_A and Ru_B sites in RuO_2 because of the oxygen atoms.
3. The opposite-spin sublattices in RuO_2 connected by $C_{4z}\mathbf{t}$ transformation.
4. The RuO_2 spin group with $\mathrm{A} = C_{4z}$ and $\mathbf{H} = mmm$.

Candidate Materials

Symmetry rules out the possibility of altermagnetism in one-dimensional (1D) chains due to the lack of rotational transformations in 1D. However, altermagnetism can manifest in two-dimensional (2D) and three dimensional (3D) crystals.

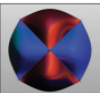
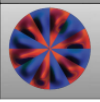
Spin-momentum locking			G	$\mathbf{R}_s^{\mathrm{III}}$	H	A	Candidate
Planar  $(K_{x/y}, K_z)$	P-2 *d*-wave	 (K_x, K_y)	*mmm*	$^{2}m^{2}m^{1}m$ (8)	2/*m*	C_{2x}	La_2CuO_4, $FeSb_2$
			4/*m*	$^{2}4/^{1}m$ (8)		C_{4z}	KRu_4O_8
	P-4 *g*-wave		4/*mmm*	$^{2}4/^{1}m^{2}m^{1}m$ (16)	*mmm*	C_{4z}	RuO_2, MnO_2, MnF_2
				$^{1}4/^{1}m^{2}m^{2}m$ (16)	4/*m*	C_{2x}	$KMnF_3$
	P-6 *i*-wave		6/*mmm*	$^{1}6/^{1}m^{2}m^{2}m$ (24)	6/*m*	C_{21}	
Bulk 	B-2 *d*-wave		2/*m*	$^{2}2/^{2}m$ (4)	$\bar{1}$	C_{2z}	CuF_2
	B-4 *g*-wave		$\bar{3}m$	$^{1}\bar{3}^{2}m$ (12)	$\bar{3}$	C_{21}	CoF_3, FeF_3, Fe_2O_3
			6/*m*	$^{2}6/^{2}m$ (12)		C_{6z}	
			6/*mmm*	$^{2}6/^{2}m^{2}m^{1}m$ (24)	$\bar{3}m$	C_{6z}	$CrSb$, $MnTe$, VNb_3S_6
	B-6 *i*-wave		*m*$\bar{3}$*m*	$^{1}m^{1}\bar{3}^{2}m$ (48)	*m*$\bar{3}$	C_{4z}	

Figure: Classification of spin-momentum locking in the third magnetic phase, highlighting planar (P) or bulk (B) characteristics, spin-group symmetries, crystallographic groups, and material candidates.[3]

Candidate Materials

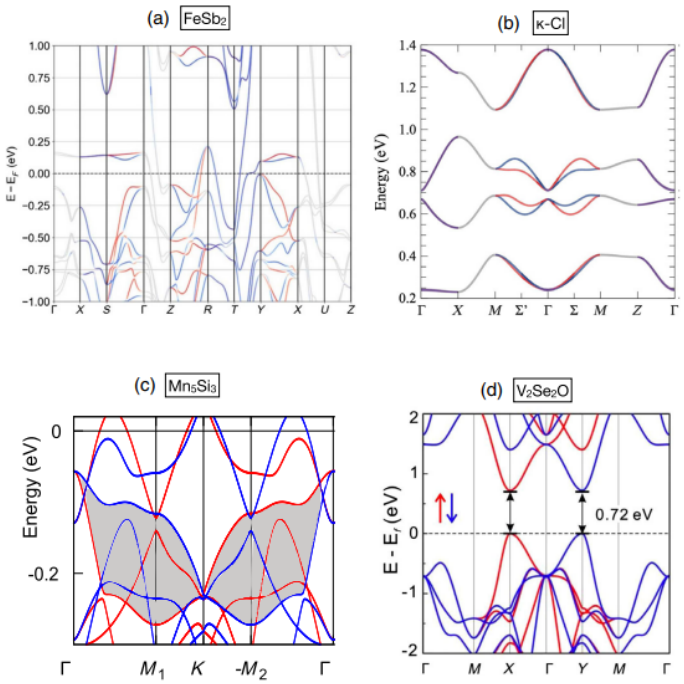


Figure: *Ab initio* spin-split band structures of depicted altermagnetic candidate materials.[2]

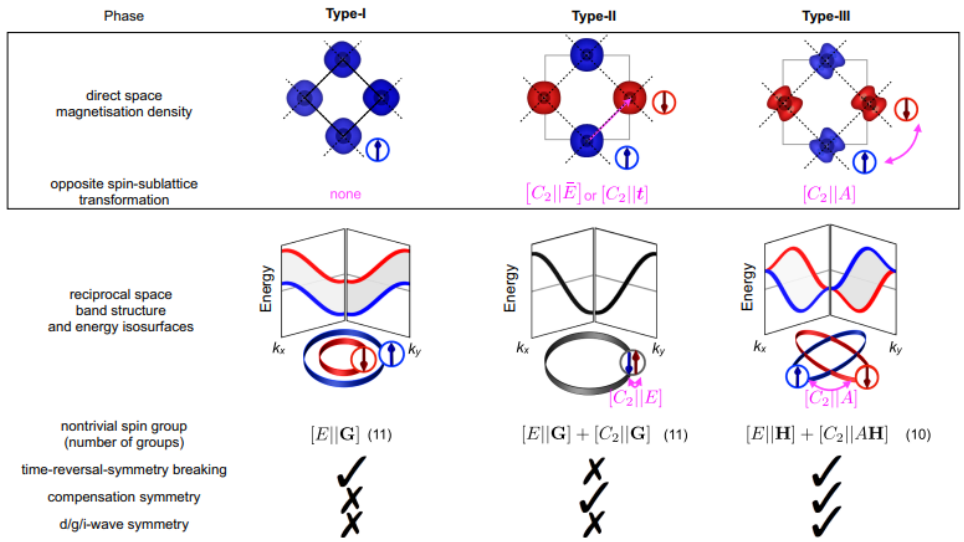


Figure: Illustration of collinear magnetic phases: ferromagnetism (Fe), antiferromagnetism (MnPt), and altermagnetism (RuO_2), showing spin arrangements, band structures, and symmetry transformations.[3]

Why CrSb?

First and most importantly, the elements (Cr and Sb) are readily available in the Raman Lab!

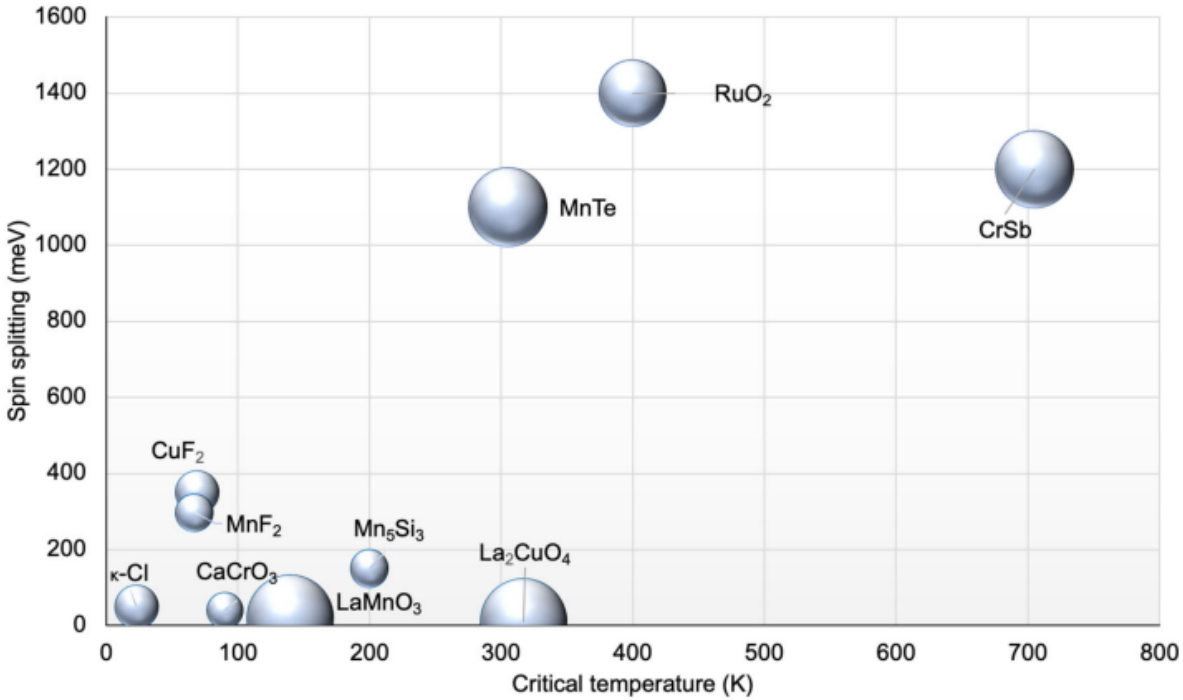


Figure: Altermagnetic candidates from ab initio calculations, shown in a transition temperature vs. spin-splitting strength diagram, with ball sizes representing the largest atomic number in the crystal.[2]

Why CrSb?

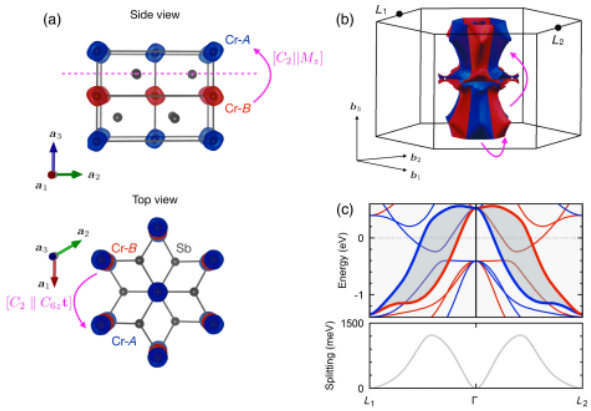


Figure: Schematic and DFT analysis of high critical temperature CrSb in the third magnetic phase, showing crystal structure with spin densities, spin-momentum locking, and band structure with spin splitting.[3]

1. It crystallizes in the hexagonal NiAs-type structure (crystal space group $P6_3/mmc$).
2. It contains the $[C_2][M_z]$ symmetry, which makes the spin-momentum locking bulklike.
3. The spin splitting is as high as 1.2 eV and it hosts an exotic spin-polarized quasiparticle which is fourfold degenerate at the Γ-point and spin split away from the Γ-point.

CrSb Thin Films

Recently, S. Reimers et al. showed unconventional band splitting near the Fermi energy in epitaxial thin films of the predicted altermagnet CrSb.[5]

- Experimental observations have yet to chart the magnitude of altermagnetic band-splitting and the precise positioning of states relative to the Fermi surface.
- In conventional antiferromagnets, spin-splitting, driven by spin-orbit coupling, reaches up to $\sim$100 meV. However, density functional theory (DFT) calculations for altermagnets predict much larger spin-splitting.
- Through a direct comparison with band structure calculations in the ordered and the unordered phase, they demonstrate a large magnitude of the spin-splitting of approximately 0.6eV near the Fermi energy.

[5] Sonka Reimers et al. "Direct observation of altermagnetic band splitting in CrSb thin films". In: *Nature Communications* 15.1 (Mar. 2024), p. 2116. ISSN: 2041-1723. DOI: 10.1038/s41467-024-46476-5

CrSb Thin Films

Epitaxial 30 nm CrSb(100) thin films were grown on GaAs(110) substrates using dc magnetron sputtering at ∼300°C, followed by in-situ annealing at ∼400°C for 15 minutes. X-ray, electron diffraction, and transmission electron microscopy confirm the films are fully epitaxial with the hexagonal c-axis of CrSb(100) aligned with the (001) direction of the substrate.

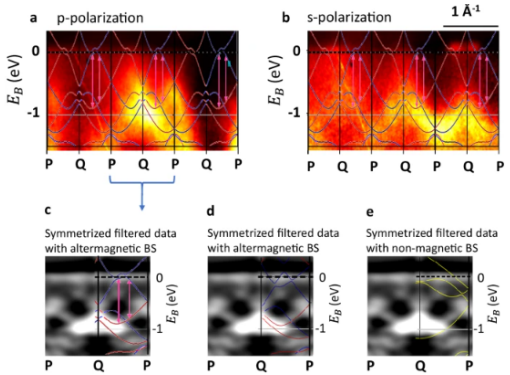


Figure: SX-ARPES intensity near the Fermi energy with spin-resolved band structure calculations for the P-Q path using p- and s-polarized photons, highlighting the importance of exchange interaction.[5]

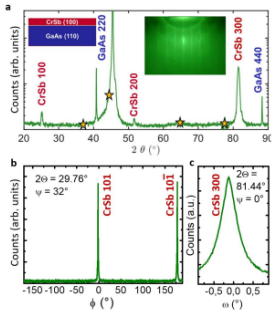


Figure: X-ray and electron diffraction of 30 nm CrSb(100) film on GaAs(110): (a) $\theta/2\theta$ scan with RHEED showing in-plane orientation; (b) ϕ-scan indicating in-plane order and 2-fold symmetry; (c) rocking curve showing 0.5° mosaicity.[5]

Motivation

Lastly, my main motivation for looking into altermagnets and, CrSb in particular is due to this interesting paper which aims to connect the worlds of alermagnetism and topology, showing that the electronic structure of the altermagnet CrSb is topological and hosts a novel Weyl semimetallic state[6].

- They observe a large momentum-dependent spin-splitting in CrSb, reaching up to 1 eV, that induces altermagnetic Weyl nodes with an associated magnetic quantum number.
- At the surface they observe their spin-polarized topological Fermi-arcs.
- This establishes that in altermagnets the large energy scale intrinsic to the spin-splitting – orders of magnitude larger than the relativistic spin-orbit coupling – creates its own realm of robust electronic topology.

[6] Cong Li et al. *Topological Weyl Altermagnetism in CrSb*. 2024. arXiv: 2405.14777

Research Direction

I wish to investigate further into this intersection, along side with working on optimizing $(Bi_xSb_{1-x})_2Te_3$ (BST) thin films for topological insulators. I will also be glad to help out on the automation tasks needed in the lab.

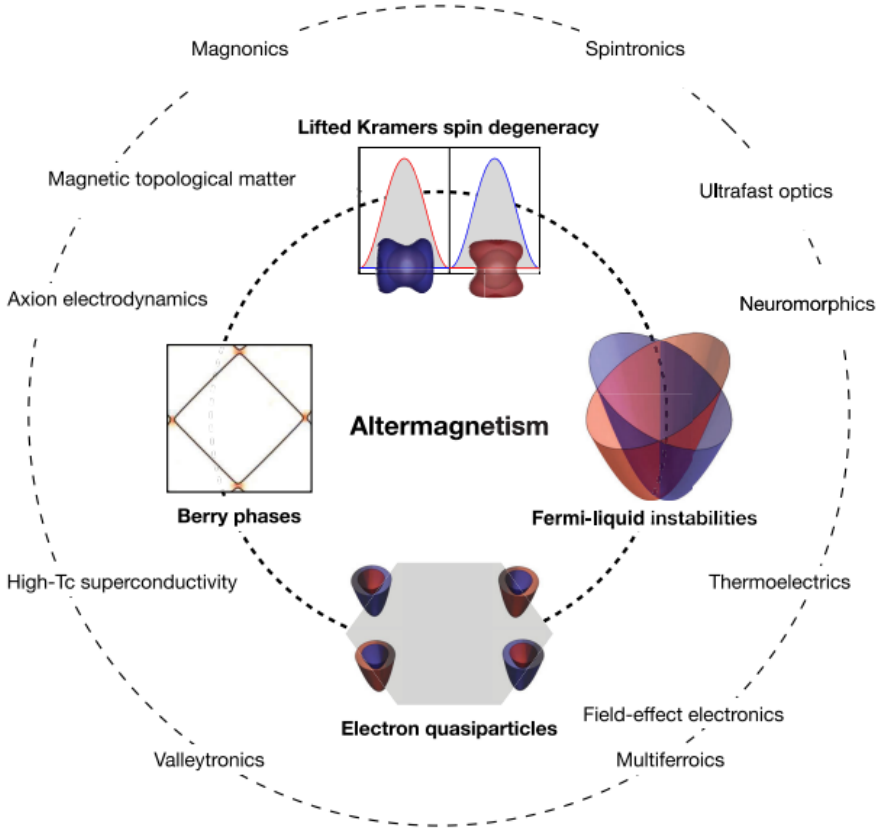


Figure: Emerging research landscape of altermagnetism.[2]

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